

Tutorial 10 Answer key

1. (a) $(11.625)_{10}$
(b) $(9.125)_{10}$
(c) $(843)_{10}$
2. (a) $(111100101)_2$
(b) $(111000011)_2$
3. 1023
4. (a) $(1001.00)_2$ or $(9)_{10}$
(b) $(100.00)_2$ or $(4)_{10}$
(c) $(10011110)_2$ or $(158)_{10}$
5. 11111000
6. -15
7. (a) $(01111)_2$
(b) $(00101)_2$
(c) $(11011)_2$ where the MSB represents sign bit
(d) $(10001)_2$ where the MSB represents sign bit
8. (a)

A	B	C	Q
0	0	0	1
0	1	1	0
1	0	1	0
1	1	1	0

(b)

A	B	C	Q
0	0	0	0
0	1	1	1
1	0	1	0
1	1	1	1

(c)

A	B	C	D	Q
0	0	1	1	0
0	1	0	0	1
1	0	0	1	1
1	1	0	0	1

9. (c)

10. (d)